EXP.NO: 6	**Write a Servlet to demonstrate session tracking using HttpSession. Implement a simple login system where the user's session is tracked.**
DATE:24/02/2026	

AIM:

To demonstrate session tracking using HttpSession and implement a simple login system where the user's session is tracked.

ALGORITHM:

1. **Input:**
   - The user fills out a web form with their name and email.
   - The form will use both GET and POST methods for data submission.

2. **Steps for Creating and Handling the HTTP GET and POST Requests:**
   1. **Create an HTML Form:**
      - Design an HTML form with two input fields: one for the name and one for the email.
      - Create two buttons to allow the user to submit the form using GET or POST methods.
      - Use the action attribute to define the Servlet URL and specify the form method (either GET or POST).
   2. **Create the Servlet to Handle GET and POST Requests:**
      - Write a Java Servlet class that will handle both GET and POST requests.
      - The servlet will have two methods:
        - doGet() to handle the GET requests.
        - doPost() to handle the POST requests.

- In both methods, retrieve the form data using request.getParameter().
- In the doGet() method, process and display the data in the URL.
- In the doPost() method, process and display the data in the HTTP request body.

3. **Handle GET Requests in the Servlet:**
    - In the doGet() method, use request.getParameter() to retrieve the name and email submitted via GET.
    - Output the data as a response, including the name and email in the URL (visible to the user).
4. **Handle POST Requests in the Servlet:**
    - In the doPost() method, use request.getParameter() to retrieve the name and email submitted via POST.
    - Output the data as a response, displaying the name and email without exposing them in the URL (more secure).
5. **Return the Output:**
    - The Servlet should return an HTML page displaying the name and email entered by the user.
    - For the GET method, the output should be visible in the URL.
    - For the POST method, the output should be visible in the response body, not the URL.
6. **Deploy and Test the Servlet:**
    - Compile the Servlet class and deploy it to a Servlet container (like Apache Tomcat).
    - Test the form using GET and POST methods to see the difference in data submission.
7. **Observe and Compare GET and POST Results:**

- Test the GET method: The form data is sent in the URL.
- Test the POST method: The form data is sent in the HTTP request body, not visible in the URL.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
<title>Dragon Stone Login</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.m
in.css" rel="stylesheet">

<style>
body {
   background: url("image.jpg") no-repeat center center/cover;
}

.overlay {
   background: rgba(0,0,0,0.85);
   height: 100vh;
}

.card {
   border-radius: 10px;
}

h3 {
   color: #ff4d4d;
}
</style>
```

```
</head>

<body>

<div class="overlay d-flex justify-content-center align-items-center">

<div class="card p-4 shadow-lg border-0" style="width: 360px;
background:#111; color:white;">

<h3 class="text-center mb-3">🍽 Dragon Stone Login</h3>

<form method="post" action="LoginServlet">

<div class="mb-3">
<label class="form-label">Username</label>
<input type="text" name="username" class="form-control" required>
</div>

<div class="mb-3">
<label class="form-label">Password</label>
<input type="password" name="password" class="form-control" required>
</div>

<button type="submit" class="btn btn-danger w-100">Login</button>

</form>

</div>

</div>

</body>
```

```
</html>
<!DOCTYPE html>
<html>
<head>
<title>Dragon Stone Dashboard</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.m
in.css" rel="stylesheet">

<style>
body {
   background: #111;
   color: white;
}
.navbar {
   background: #000;
}
.card {
   background: #222;
   color: white;
}
</style>
</head>

<body>

<nav class="navbar px-3">
<span class="navbar-brand text-danger fw-bold">🍽 Dragon Stone</span>

<span>
Welcome, <strong><%= session.getAttribute("username") %></strong>
</span>
```

```html
<a href="LoginServlet?action=logout" class="btn btn-outline-danger btn-sm">Logout</a>
</nav>

<div class="container mt-4">

<h3 class="mb-4">Available Tables</h3>

<div class="row g-3">

<div class="col-md-4">
<div class="card p-3">
<h5>Table 1</h5>
<p class="text-success">2 Seats - Available</p>
<a href="booking.html" class="btn btn-danger">Book Now</a>
</div>
</div>

<div class="col-md-4">
<div class="card p-3">
<h5>Table 2</h5>
<p class="text-success">4 Seats - Available</p>
<a href="booking.html" class="btn btn-danger">Book Now</a>
</div>
</div>

<div class="col-md-4">
<div class="card p-3">
<h5>Table 3</h5>
<p class="text-danger">6 Seats - Booked</p>
<button class="btn btn-secondary" disabled>Booked</button>
```

</div>

</div>

</div>

</div>

</body>

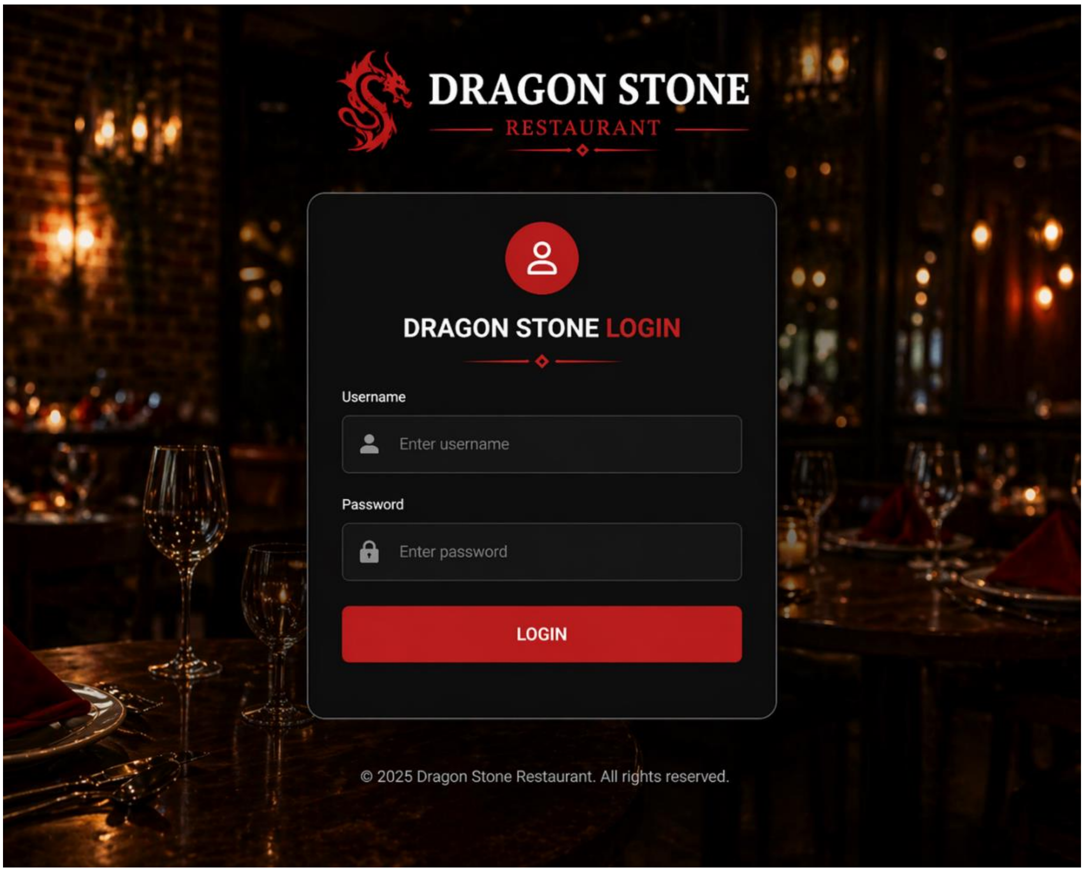
</html>

Servlet6.java:

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response)
            throws ServletException, IOException {

        String action = request.getParameter("action");

        // Logout
        if ("logout".equals(action)) {
            HttpSession session = request.getSession(false);
            if (session != null) {
                session.invalidate();
            }
            response.sendRedirect("login.html");
            return;
        }

        // Session check
        HttpSession session = request.getSession(false);
        if (session != null && session.getAttribute("username") != null) {
            request.getRequestDispatcher("dashboard.jsp").forward(request,
response);
        } else {
```

```
            response.sendRedirect("login.html");
        }
    }

    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
            throws ServletException, IOException {

        String username = request.getParameter("username");
        String password = request.getParameter("password");

        // Simple login
        if ("admin".equals(username) && "1234".equals(password)) {

            HttpSession session = request.getSession();
            session.setAttribute("username", username);
            session.setMaxInactiveInterval(300);

            response.sendRedirect("LoginServlet");

        } else {
            response.getWriter().println("<h3 style='color:red;text-
align:center;'>Invalid Login</h3>");
        }
    }
}
```

Output:


DRAGON STONE
RESTAURANT
DRAGON STONE LOGIN
Username
Enter username
Password
Enter password
LOGIN
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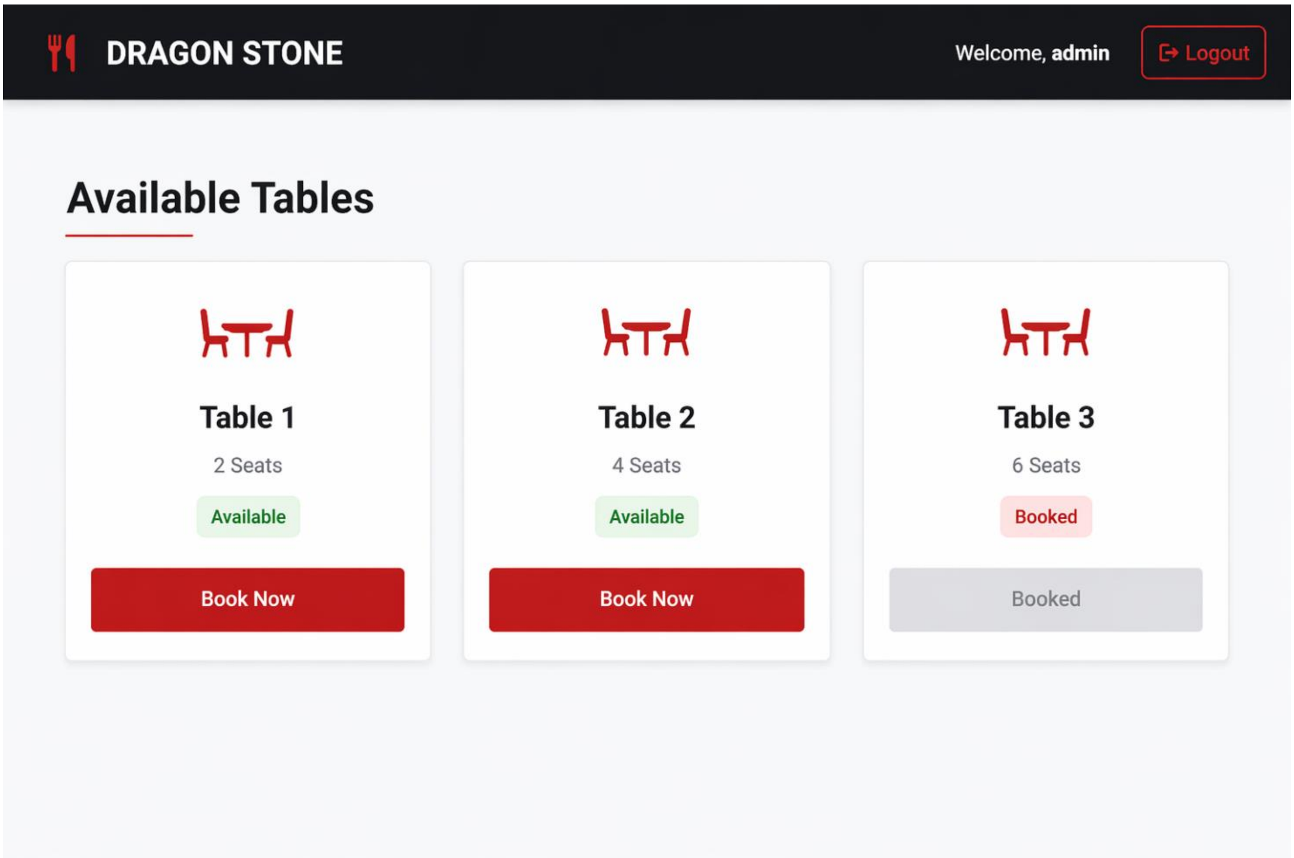
DRAGON STONE
Welcome, admin
Logout
Available Tables
Table 1
2 Seats
Available
Book Now
Table 2
4 Seats
Available
Book Now
Table 3
6 Seats
Booked
Booked

RESULT:

A Servlet is created to demonstrate session tracking using HttpSession and implemented a simple login system where the user's session is tracked.